MEHMET A. GULER, Ph.D.

Senior Mechanical Integrity / FEA Engineer — Structural Analysis | Fatigue | Failure Assessment

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PROFESSIONAL SUMMARY

Senior mechanical integrity and computational mechanics engineer with extensive industrial, consulting, and R&D experience in **finite element analysis, structural stress analysis, fatigue, fracture mechanics, failure investigation, nonlinear analysis, and thermal-structural analysis**. Former **Senior Structural & Thermal Engineer at CD-adapco (now Siemens)** with hands-on **ANSYS** experience analyzing industrial equipment, power-generation, gas-turbine, compressor, and automotive components. Experienced in translating engineering problems into computational models, evaluating structural integrity and failure-critical regions, **correlating simulations with physical measurements and tests**, and providing technical recommendations for design improvement, reliability, and safety.

TECHNICAL EXPERTISE

Mechanical Integrity / Structural Assessment: stress analysis, structural integrity, fatigue and durability, fracture mechanics, failure investigation, material behavior, welded structures, design evaluation
FEA / Computational Analysis: ANSYS, LS-DYNA, Abaqus, ANSA; linear and nonlinear FEA, stress/strain analysis, contact and friction, material nonlinearity, large deformation, buckling, model verification and validation
Thermal / Dynamic Analysis: thermal and coupled thermal-structural analysis, thermomechanical loading, structural dynamics, modal analysis, vibration, transient response
Testing / Validation / Reporting: strain-gauge measurements, static and fatigue testing, experimental correlation, parametric studies, design optimization, simulation automation, engineering calculations, technical reports

RELEVANT INDUSTRIAL ENGINEERING EXPERIENCE

CD-adapco Inc. (now Siemens) — **Melville, NY**
Senior Structural & Thermal Engineer — Sep 2000–Sep 2004

- Developed and delivered **structural, stress, thermal, dynamic, nonlinear, and coupled thermomechanical FEA** for industrial equipment, power-generation, gas-turbine, compressor, engine, automotive, and heavy-equipment applications.
- Performed 3D stress and dynamic analysis of **gas-turbine and compressor components** under combined structural, thermal, and dynamic loading using **ANSYS**.
- Performed **structural stress analysis of a glass-lined industrial reactor**, developing the FE model and evaluating structural response and critical stress regions.
- Conducted thermal analysis of diesel-engine components and coupled thermal-structural analysis of **EGR coolers**; identified crack-prone regions and recommended design modifications.
- Developed nonlinear FE models incorporating **contact, friction, nonlinear material behavior, large deformation, and thermal-structural loading**.
- Investigated component failures and translated physical behavior, loading conditions, materials, and engineering requirements into appropriate computational models and analysis strategies.
- Developed engineering programs and scripts for FE model generation, analysis setup, parametric studies, automated processing, and verification.

GE Marmara Technology Center — Engineering Consultant — 2010
Performed **ANSYS finite element stress analysis of aircraft-engine components**, evaluating stresses and structural response of high-performance turbomachinery hardware.

INDUSTRY CONSULTING & FATIGUE ASSESSMENT

SANKO / MST Construction Equipment — Engineering Consultant — 2016–2018
Developed **structural FEA and fatigue-life models** for 7–18 m welded telehandler booms; identified fatigue-critical regions using Eurocode 3, BS 7608, IIW, and FKM methodologies and **correlated computational predictions with strain-gauge measurements and physical static/fatigue tests** for model validation, life assessment, and structural design improvement.

ADDITIONAL INDUSTRIAL ENGINEERING EXPERIENCE

TEMSA Global Inc. — **Adana, Türkiye**
Chief Structural Analysis Engineer — Mar 2005–Sep 2006

- Led structural CAE activities involving finite element model development, nonlinear structural analysis, structural dynamics, failure evaluation, verification, and simulation-driven design.
- Developed full-vehicle FE models using **ANSA** and performed nonlinear structural/crash simulations using **LS-DYNA**.
- Performed 3D structural and dynamic analyses using **ANSYS**; identified critical regions, assessed structural integrity, and recommended design modifications.
- Conducted modal analyses of multiple vehicle platforms and evaluated structural vibration modes.

INDUSTRY CONSULTING & APPLIED ENGINEERING

Arçelik Dishwasher Inc. — Project Consultant / Principal Investigator 2013–2015
Led a $135K industry-funded FEA program involving structural analysis, simulation-driven design, and packaging optimization.

Oyak Renault Inc. — Project Consultant 2011–2013
Performed structural and dynamic FEA, side-crash and frontal-crash analysis, and energy-absorber development; evaluated structural response and design alternatives under severe loading.

TEMSA Global Inc. — Project Consultant / Principal Investigator 2008–2010
Led a $225K government/industry R&D program developing energy-absorbing structural systems using nonlinear FEA, impact analysis, optimization, and simulation-driven design.

ENGINEERING & R&D LEADERSHIP

American University of the Middle East (AUM) — **Kuwait**
Professor of Mechanical Engineering — Sep 2018–Present

- Lead computational engineering R&D in **finite element analysis, structural mechanics, nonlinear mechanics, fatigue, fracture, contact, structural dynamics, composites, and optimization**.
- Develop analytical and numerical models to investigate structural response, damage and failure mechanisms; perform verification and validation and mentor researchers in computational mechanics and FEA.

TOBB University of Economics and Technology — **Ankara, Türkiye**
Professor / Associate Professor / Assistant Professor — Sep 2006–Sep 2018

- Led industry-sponsored engineering programs involving **nonlinear FEA, fatigue, fracture, structural dynamics, crashworthiness, optimization, composites, and contact mechanics**.
- Directed computational and experimental programs with automotive, heavy-equipment, transportation, and consumer-product partners and supervised multidisciplinary engineering teams.

SELECTED MECHANICAL INTEGRITY & VALIDATION EXPERIENCE

- **Industrial Reactor Structural Analysis — CD-adapco:** Developed an FE model of a glass-lined reactor and performed structural stress analysis to evaluate component response and critical stress regions.
- **Fatigue-Life Assessment — SANKO/MST:** Developed structural FEA and fatigue-life assessments for welded components and correlated predictions with strain-gauge measurements and physical static/fatigue tests.
- **Failure Investigation / Thermal-Structural Analysis:** Evaluated crack-prone and highly stressed regions in engine and industrial components using nonlinear and coupled thermomechanical FEA and recommended design modifications.
- **Industrial Component Analysis — CD-adapco / GE:** Performed ANSYS stress and dynamic analyses of turbomachinery and aircraft-engine components under structural, thermal, and dynamic loading.
- **Fracture / Damage Mechanics Research:** Developed computational and analytical models involving fracture, damage evolution, contact, nonlinear material response, and structural failure.

EDUCATION

Ph.D., Mechanical Engineering, Lehigh University, Bethlehem, PA
Dissertation: *Contact Mechanics of FGM Coatings*

M.S., Mechanical Engineering, Lehigh University, Bethlehem, PA

M.S., Computer Technology in Manufacture, University of Sussex, UK

B.S., Mechanical Engineering, Middle East Technical University, Türkiye